

CHETAN KHARADE

AI/ML Engineer | Generative AI & LLM Engineer | AI Agent Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building, training, fine-tuning, and deploying AI systems across Generative AI, Large Language Models (LLMs), AI agents, and computer vision. Skilled in developing production backend services with FastAPI, integrating vector databases and Retrieval-Augmented Generation (RAG) pipelines, and self-hosting and deploying AI models on Linux servers and cloud infrastructure (Docker, Google Cloud Run, Modal). Experienced across the full AI production lifecycle, from model training and fine-tuning through API development, deployment, and ongoing production maintenance.

PROFESSIONAL EXPERIENCE

AI Developer | Helios Infotech — Pune, India

April 2025 – Present

- Design, develop, and deploy AI/ML applications spanning Generative AI, LLMs, AI agents, and computer vision, covering the full lifecycle from model development through production deployment.
- Build and fine-tune machine learning and deep learning models, including self-hosted LLM deployment for production use cases.
- Develop backend services and REST APIs using FastAPI to serve AI models and integrate them into production applications.
- Architect AI agent systems with multi-agent routing, session and state management, streaming responses, and tool calling for conversational and task-based applications.
- Containerize and deploy AI applications using Docker, managing production workloads on Google Cloud Run and Modal.
- Integrate voice AI and real-time communication capabilities using LiveKit and WebRTC to build self-hosted voice-agent infrastructure.
- Administer Linux servers for self-hosted AI model deployment, including configuration, monitoring, and troubleshooting of production systems.
- Automate workflows and maintain production AI systems, ensuring reliability across model serving, APIs, and database integrations.

TECHNICAL SKILLS

- **AI / Machine Learning:** Python, Scikit-learn, TensorFlow, Keras, Machine Learning, Deep Learning, NLP, Computer Vision, OCR, Object Detection
- **Generative AI / LLM:** LLMs, LLM Fine-Tuning, Generative AI, Multimodal AI, AI Agents, RAG, Prompt Engineering, OpenAI, LangChain, LangGraph, Embeddings, Vector Search
- **Backend:** FastAPI, REST APIs, AsyncIO, Uvicorn, Pydantic
- **Databases / Storage:** PostgreSQL, MongoDB, SQLite, Qdrant, AWS S3
- **Computer Vision:** OpenCV, Real-ESRGAN, Reti-Diff, Image Processing, RAW Image Processing, HEIC Processing
- **Cloud / MLOps / Deployment:** Docker, Linux, Modal, Google Cloud Run, AWS, Model Serving, Self-Hosted Models, Server Administration, Production Deployment, Cron
- **Voice / Real-Time AI:** LiveKit, WebRTC, Speech-to-Text, Text-to-Speech, Voice Agents
- **Developer Tools:** Git, GitHub, Jupyter Notebook, VS Code, PyCharm, Google Colab

SELECTED AI/ML PROJECTS

DigitalWardrobeAI — Multimodal AI Digital Wardrobe Platform

- Built an AI-powered digital wardrobe platform with automatic background removal, clothing classification, and vision AI metadata extraction covering category, color, material, season, and style.
- Implemented embeddings-based semantic search with Qdrant, enabling natural-language wardrobe queries and AI-driven outfit generation.
- Integrated OpenAI and Google GenAI models with a FastAPI backend, MongoDB, and AWS S3 for image storage and retrieval.
- Developed virtual try-on functionality combining computer vision and generative AI.

Tech: Python, FastAPI, Gradio, OpenAI, Google GenAI, Qdrant, MongoDB, AWS S3, rembg, ONNX Runtime

PLUMBWISE (DIYplumbing) — AI Agent Plumbing Assistant

- Developed a conversational AI agent for plumbing troubleshooting using LangChain and LangGraph, with tool calling and safety-first escalation for high-risk situations.
- Built a RAG pipeline for document-grounded responses and integrated web search and image analysis for multimodal troubleshooting.
- Implemented real-time voice interaction using LiveKit and WebRTC with STT/TTS integrations.
- Engineered stateful, streaming chat threads on a FastAPI backend backed by MongoDB (Motor/PyMongo) and AWS S3.

Tech: Python, FastAPI, OpenAI, LangChain, LangGraph, MongoDB, LiveKit, WebRTC

Perfusion Mind AI — Multi-Agent LLM Application

- Developed a multi-agent LLM application with dynamic assistant routing using OpenAI Assistants, session-based state management, and streaming responses.
- Implemented tool calling and run cancellation for controllable, production-grade agent behavior.
- Containerized the application with Docker and deployed it to Google Cloud Run with automated deployment and production logging.

Tech: Python, FastAPI, OpenAI Assistants, Chainlit, Docker, Google Cloud Run

NewsAggregator — Production RSS News Aggregation & Search Platform

- Built a production-grade async news aggregation platform covering 7 Indian publishers, including 207+ RSS feeds for Indian Express.
- Designed an incremental scraping system with a watermarking mechanism and multi-pass deduplication, plus author enrichment and URL extraction.
- Developed an AST-based boolean and fuzzy-match search query engine with cross-lingual English–Hindi search on FastAPI, backed by PostgreSQL and SQLite.
- Automated feed orchestration with cron scheduling and deployed the system on a Linux server.

Tech: Python, asyncio, curl_cffi, FastAPI, PostgreSQL, SQLite, BeautifulSoup, Git, Cron

ImageEnhancement — Cloud-Based AI Image Enhancement System

- Built a cloud-based image enhancement system supporting JPG, PNG, and RAW formats including ARW, CR2, DNG, and HEIC.
- Implemented AI super-resolution with up to 4x upscaling using Real-ESRGAN and Reti-Diff on NVIDIA L40S GPUs.
- Deployed GPU-backed inference on Modal with a FastAPI/Gradio interface, delivering low-light enhancement, contrast correction, and white balance correction.

Tech: Python, Modal, FastAPI, Gradio, Real-ESRGAN, Reti-Diff, OpenCV, rawpy, pillow-heif, NVIDIA L40S GPU

ADDITIONAL PROJECTS

- **Greyhound Racing Prediction** — Trained a machine learning model on historical racing performance data with feature engineering for race outcome prediction.
- **CCTV Person & Number Plate Detection** — Contributed to a computer-vision solution for person and number-plate detection used in a company project associated with a Gujarat cybersecurity initiative.
- **AI Children's Storybook Generator** — Built a generative AI application producing personalized children's storybooks for readers aged approximately 1–12 years.
- **Invita Watches Sales Agent** — Developed a backend AI agent to assist salespeople with guided customer conversations.
- **Self-Hosted LiveKit Voice Agent Infrastructure** — Self-hosted a LiveKit voice-agent platform for real-time voice AI workflows over WebRTC.

EDUCATION

B.E., Information Technology Engineering — Savitribai Phule Pune University

January 2021 – June 2025 | CGPA: 8.60

HSC, Science — Samarth Junior College, Belhe

2019 – 2021 | 79.50%